

RPM COUNTER USING MICROCONTROLLER AT89C4051

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Counting the revolutions per minute (RPM) of motors—determining the motor speed—is essential in the field of industrial automation. It is useful especially for closed-loop control systems where proper action can be taken in case the actual RPM deviates from the set RPM.

Here is a project based on microcontroller AT89C4051 that measures

and shows on an LCD the RPM of a running motor. Using a proper transducer, first the rotations of the motor are converted into pulses. The generated pulses are counted by the microcontroller for a fixed time (say, one second). The count is multiplied by a factor to get the exact RPM and then displayed; if time is one second, the factor is 60.

Circuit description

Fig. 1 shows the block diagram of the RPM counter based on AT89C4051 microcontroller that generates pulses for every rotation of the motor, counts them and shows on the LCD.

On a fixed base, there is a laser source on one side and the combination of a light-dependent resistor (LDR), pulse generator, microcontroller and LCD on the other side. Both the

arrangements are housed in separate wooden cabinets such that the laser beam falls directly on the LDR. The motor is placed on top of the laser source. A slotted wheel is attached to the motor shaft. The wheel is so big that it can interrupt the laser beam falling on the LDR.

As the motor rotates, the slotted wheel also rotates. The laser beam falls on the LDR whenever the slot aligns with the laser beam and LDR, producing one pulse. Thus the rotations of the motor are converted into pulses that can be counted by the program in the microcontroller. Pulses are counted for one second. The pulse count is multiplied by 60 (because 1 RPM = 60 RPS) and finally shown on the LCD.

Fig. 2 shows the circuit of the RPM meter. It comprises microcontroller

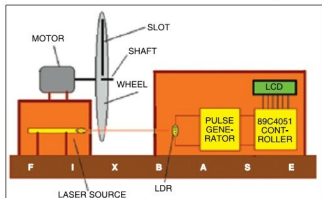


Fig. 1: Block diagram of the RPM counter based on microcontroller AT89C4051

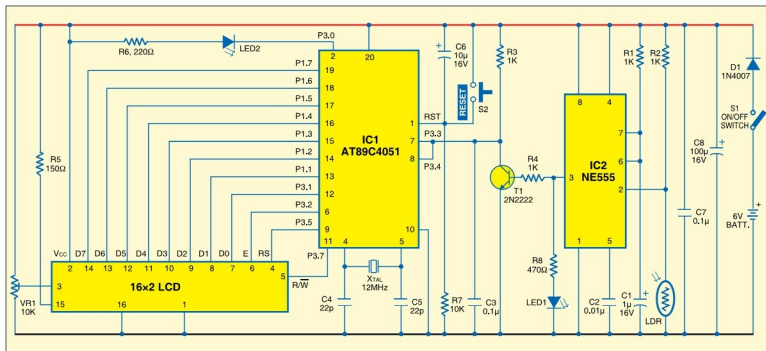


Fig. 2: Circuit of the RPM counter using AT89C4051